Hole and Vacancy Statistics of Equilibrium Hard-Sphere Crystals

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1. Motivation

The "hole" statistics, a.k.a. nearest-neighbor distribution functions of a statistically homogeneous many-body systems can be expressed as integrals of all n-point correlation functions $g_2, g_3, \ldots$ (Torquato, Lu and Rubinstein, Phys. Rev. A. 1990).
We define the following "hole" statistics:

- $H_V(r)dr$ = **probability that at an arbitrary located point in the system, the nearest point in the point process lies between r and $r + dr$.**
- $E_V(r) = \int_r^\infty H_V(r')dr'$: **probability that given an arbitrary location in the void, a ball of radius r centered at that location is devoid of points**
- $\rho s_1(r) G_V(r)dr$ = **probability of finding a particle between rand $r + dr$ given that one has found a hole of radius r.**

They have important physical significance. For example:

- **The pressure of an equilibrium hard-sphere (HS) system in $\mathbb{R}^d$ at packing fraction is given by**

$$\frac{p}{\rho k_B T} = 1 + 2^d \phi g_2(D^+) = 1 + 2^d \phi G_V(D). \tag{1}$$

- **The chemical potential of an equilibrium HS system is given by**

$$\mu = \mu_{\mathbf{id}} - k_B T \ln(E_V(D)) \tag{2}$$

- **The "void" probability density function for crystals can be used to estimate the concentration of point defects (Pronk and Frenkel, J. Chem. Phys. 2001).**
- **Difference in $G_V(r)$ between nonequilibrium and equilibrium systems with identical pair statistics defines a nonequilibrium index Γ (Wang and Torquato, Soft Matter. 2023).**

Significance of this work

- Sampling of extremely rare events is of great interest, as they are related to the dynamics of phase transitions, glass formation, etc.
- The standard method for computing the void statistics becomes inaccurate for holes such that $E_V(r) < 10^{-10}$.
- The hole statistics for hard-sphere liquids is very well known (Torquato, Lu and Rubinstein, PRA 1990), but knowledge for the crystal phase is very limited.
- Simulations of crystals are typically done in the canonical ensemble or NPT ensemble. However, to study hole statistics in crystals, it is desirable to simulate in the grand canonical (GC) ensemble, where particles can be added and removed to create and migrate vacancies.
- However, particle addition moves in the standard GCMC simulations for crystals have a very high rejection rate, making such simulations impractical.
- One needs a new accurate algorithm to compute hole statistics based on biased sampling in the grand canonical ensemble.

2. Objectives

We study the following statistical functions that describe local free spaces in crystals:

- **The nearest-neighbor distribution $H_V(r), E_V(r), G_V(r)$. We devise an algorithm for computing statistics of rare holes that is much more accurate than the standard method, and accurately compute $G_V(r)$ for 2D and 3D hard-sphere crystals up to a distance beyond the sphere diameter.**
- **The vacancy concentration X. We study this quantity via theory and simulation.**
- **The local packing fraction for the first coordination shell ϕ_1. We study how this quantity predicts phase transitions.**

3. Standard methodology to compute hole statistics

- **The standard methodology for computing the "void" hole statistics samples holes by adding n random test points $T = \{\mathbf{t}_1, \mathbf{t}_2, ..., \mathbf{t}_n\}$ into the particle configurations $\mathbf{r}^N$.**
- **For each test point $\mathbf{t}_i$, compute its distance d_i to its nearest particle in $\mathbf{r}^N$ in periodic boundary conditions.**
- **$E_V(r)$ is then computed as**

$$E_V(r) = \frac{|\mathbf{t}_i \in T : d_i > r|}{n}. \tag{3}$$

  The result for $E_V(r)$ is averaged over s independent configurations. $H_V(r)$ and $G_V(r)$ can be easily derived from $E_V(r)$.
- **For $n/N = 10^6$ and $Ns = 10^5$ (the typical limit of computational power), this method becomes inaccurate (error in $G_V(r) > 100\%$) for $E_V(r) < 10^{-10}$, and is unable to find holes such that $E_V(r) < 10^{-15}$.**
- **Need biased sampling to sample large holes.**

4. Biased sampling methodology

4.1 Biasing potential

- **We utilize a fictitious biasing potential u that encourages large holes as follows:**
- **Let t be a single random test point in the configuration. If no biasing potential is present, the distribution $p_0(r)$ of distances between t and its nearest neighbor in $\mathbf{r}^N$ is simply $H_V(r)$.**
- **The biasing potential $u(r)$ (where r is the nearest-neighbor distance from t to particles in $\mathbf{r}^N$) is designed such that it is roughly proportional to $\ln[p_0(r)] = \ln[H_V(r)]$, so that one can observe larger holes centered at t.**
- **The form of $u(r)$ is obtained from a fit of $ln[H_V(r)]$ in the range where $H_V(r)$ can be accurately computed via the standard method ($0 \leq r/D \leq 0.7$). For 1D hard rods at $\phi = 0.95$, $u(r)$ is given by**

$$u(r) \approx \ln[H_V(r)] = \begin{cases} 0.461, & r < 0.5D \\ -38.089r/D + 19.505, & r \geq 0.5D \end{cases} \tag{4}$$

4.1 Biasing potential

- **For 2D hard disks at $\phi = 0.73$, The initial $u(r)$ is given by**

$$u_0(r) = \begin{cases} \ln(1.858\pi r/D) - \ln(0.929\pi) - 20, & r < 0.5D \\ -40r/D, & r \geq 0.5D. \end{cases} \tag{5}$$

  It is updated in each iteration as $u_{i+1}(r) = u_i(r) + \ln[p_i(r)]$, where $p_i(r)$ is the equilibrium nearest neighbor distance distribution to the test point under $u_i(r)$.

- **The biasing potential (4) is similar to the linear potential used in [Zhang, Torquato and Stillinger, Soft Matter, 2017], given by**

$$F(r_{ic}, R_f) = \begin{cases} \frac{R_f}{r_{ic}} - 1, & r_{ic} < R_F \\ 0, & r \geq R_F, \end{cases} \tag{6}$$

  where r_{ic} is the distance between a particle to the center of the simulation box, and R_F is the radius of the hole that one attempts to add to the configuration.

- **The difference between (4) and (6) is that (4) is informed directly by the form of $H_V(r)$, as it attempts to "flatten" the nearest-neighbor distance distribution, whereas the objective of using (6) is to find the largest hole.**

4.2 "Centered" grand canonical ensemble

The biased sampling scheme enables one to address the problem of high rejection rate in the standard GCMC simulations: New particles inserted in the vicinity of the test point have a higher probability to be accepted. We propose the "centered" grand canonical ensemble, defined by the following moves that change N:

- **Particle addition: Randomly add a particle whose center is in the spherical region of radius $\lambda - D$ from the test point, where λ is the nearest neighbor distance to the test point before addition.**
  - **This ensures that the newly added particle becomes the nearest neighbor to the test point.**
  - **Do nothing if $\lambda < D$.**
- **Particle deletion: Remove the nearest actual particle from the test point if the second nearest neighbor $\lambda' > D$ and $\lambda < \lambda' - D$.**
  - **Do nothing if $\lambda' \leq D$.**

4.2 "Centered" grand canonical ensemble

- **To maintain detail balance, the addition and deletion moves are accepted with probabilities**

$$P_{\mathbf{add}} = \min\left[1, \exp(-\beta\Delta\Phi + \beta\mu')v_1(\lambda - D)\right]$$
$$P_{\mathbf{del}} = \min\left[1, \frac{\exp(-\beta\Delta\Phi - \beta\mu')}{v_1(\lambda' - D)}\right] \tag{7}$$

  where $\Delta\Phi$ is the change in potential energy caused by the move, $\mu' = \mu - k_B T \ln \Lambda^d$ is the effective chemical potential, and λ' is the second nearest neighbor distance before the attempted deletion.
- **Because the system is not homogeneous immediately after a particle addition or removal, an addition or deletion is attempted every 10 sweeps of particle displacement.**
- **For computing small-hole statistics, one can speed up the simulation by moving particles near the test point more frequently (while maintaining detail balance). However, for holes with $r \geq D$, doing this leads to slow equilibration.**

4.3 Simulations

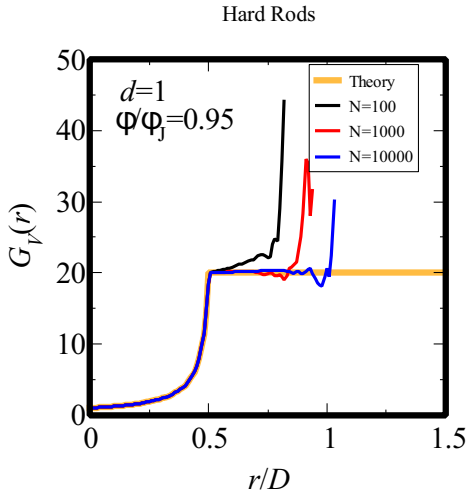
- **Starting from a perfect crystal, MC simulations are performed for N_0 d-dimensional hard spheres with one single random test point. The potential between the test point and its nearest particles is given by a fit of $\ln[H_V(r)]$. Both the particles and the test point undergo MC moves.**
- **To speed up the simulation, we first let the biasing potential $u(r)$ act on M nearest particles to create a large hole. After the nearest-neighbor distribution stabilizes, $u(r)$ then acts on the nearest particle only; $M = 2, 6$ for $d = 1, 2$, respectively.**
- **After equilibrium is reached, the distribution of nearest-neighbor distance from the test point to the particles is sampled for 800,000 sweeps and averaged over 200 trajectories. This biased distribution is labeled $p_i(r)$.**
- **One can recover $H_V(r)$ by folding back the distribution:**

$$H_V(r) \propto p_i(r) \exp[u(r)]. \tag{8}$$

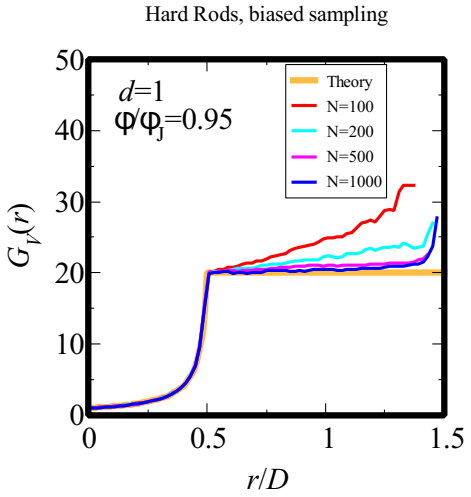
  Normalization is required to ensure that $\int_0^\infty H_V(r)dr = 1$.
- **Repeat until $p_i(r)$ is a uniform distribution on $[0, 1.5D]$.**

5. Results for biased sampling

5.1 Canonical ensemble results for 1D hard rods at $\phi = 0.95$



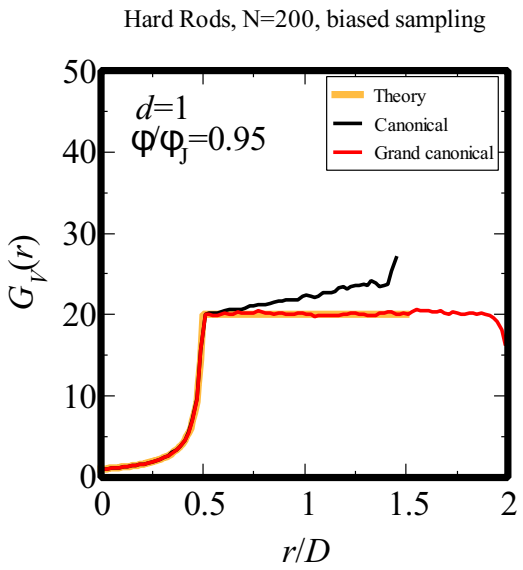
The conditional hole probability function $G_V(r)$ computed via the standard method in the canonical ensemble.



The conditional hole probability $G_V(r)$ computed with biased sampling in the canonical ensemble.

- **The biased sampling scheme is able to compute the hole statistics over a much larger range than the standard method.**
- **The simulated $G_V(r)$ via biased sampling approaches the theoretical infinite-size limit as N increases.**
- **For $N = 1000$, $G_V(D) = 20.5$, which deviates just 2.5% from the theoretical value 20.**
- **By contrast, the standard method requires $N \geq 10000$ to be able to sample $G_V(D)$ accurately.**

5.2 Grand canonical ensemble results for 1D hard rods at $\phi = 0.95$



The conditional hole probability $G_V(r)$ computed with biased sampling.

- **For $N = 200$, $G_V(D) = 20.1$ in the grand canonical ensemble, which deviates just 0.5% from the theoretical value.**
- **In fact, $G_V(r)$ in the grand canonical ensemble is accurate within 3% up to $r = 1.9D$, where $E_V(r) \sim 10^{-25}$.**
- **By contrast, for $N = 200$ in the canonical ensemble, $G_V(D) = 22.3$, which deviates 10% from the theoretical value.**
- **Thus, hole statistics in the grand canonical ensemble approaches the infinite-size limit much faster than those in the canonical ensemble. This is due to the former's ability to create and migrate vacancies.**

5.3 Grand canonical ensemble for 2D hard disks at $\phi = 0.73$

- **We simulate a triangle lattice at packing fraction just above the hexatic phase $\phi = 0.73$ with $N = 400$ in a rhombic simulation box.**

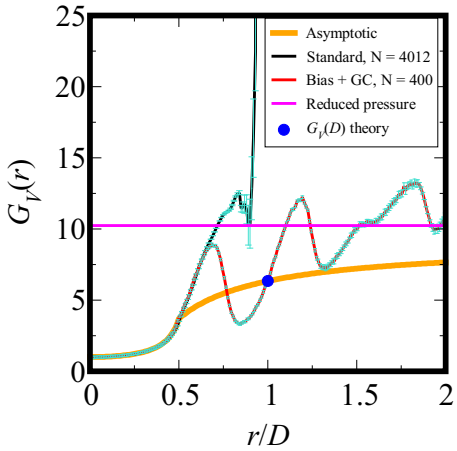
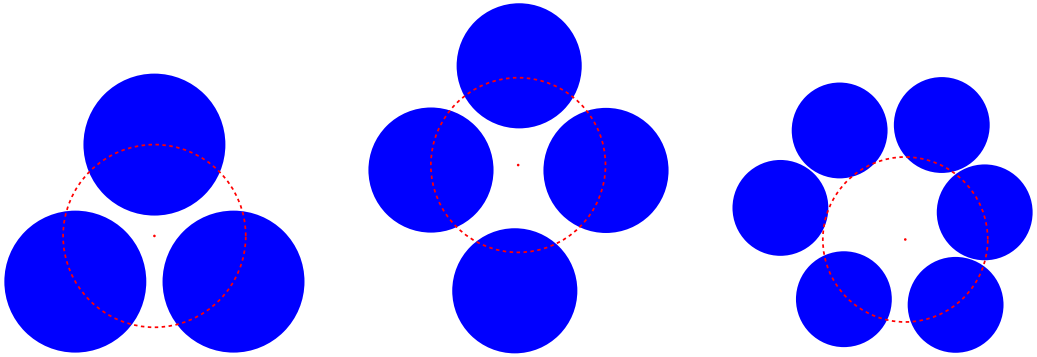


Figure: $G_V(r)$ for 2D hard disk crystal at $\phi = 0.73$, $\phi/\phi_J = 0.805$. The biased sampling algorithm yields much smaller error bars compared to the standard method. The standard method also has significant finite size effect because it uses the canonical ensemble.

- **Biased sampling in the grand canonical ensemble shows that $G_V(r)$ has the following extrema:**
  - **Maxima:** $0.70D, 1.21D, 1.87D$
  - **Minima:** $0.85D, 1.34D, 1.92$
  - **Shoulder at** $1.61D$.
- **The oscillations persist for a large range of r. The oscillation period is** $0.5D$.
- $G_V(r)$ **has an oscillatory tail that asymptotes to the reduced pressure** $p/(\rho k_B T) = 10.25$.

5.4 Local geometry in the vicinity of holes

We consider the triangle-lattice crystal at $\phi = 0.73$.



Triangular hole the perfect triangle lattice with radius $r = 0.64D$.

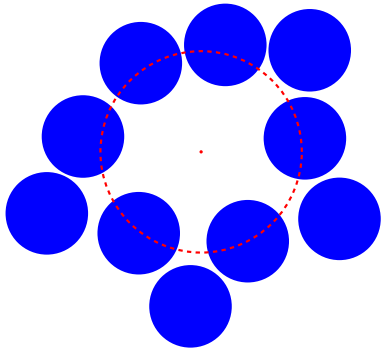
Local configuration in the vicinity of a hole of radius $r = 0.70D$, corresponding to a local maximum in $G_V(r)$.

Local configuration in the vicinity of a hole of radius $r = 0.85D$, corresponding to a local minimum in $G_V(r)$.

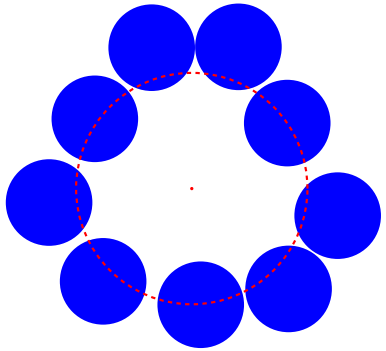
- **The "coordination number" of holes Z, i.e., the number of actual particles that overlap with the hole, increases with the hole radius.**
- **The maximum in $G_V(r)$ at $r = 0.70D$ corresponds to $Z = 4$, i.e., a sheared unit cell.**
- **The minimum in $G_V(r)$ at $r = 0.85D$ corresponds to $Z = 6$.**
- **Intuitively, since bond angles $\pi/3$ and $2\pi/3$ dominate triangle-lattice configurations, triangular and hexagonal holes are less "strained" than square holes.**

5.4 Local geometry in the vicinity of holes

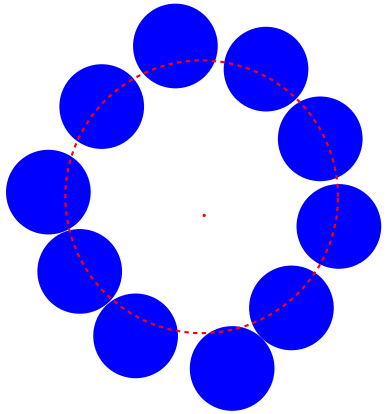
We consider the triangle-lattice crystal at $\phi = 0.73$.



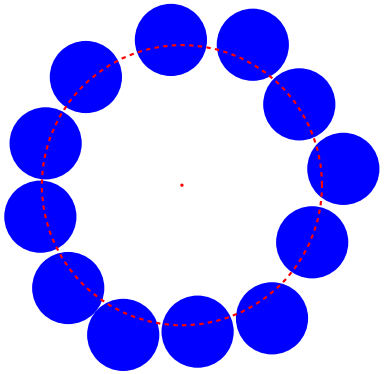
Local configuration in the vicinity of a hole of radius $r = 1.21D$, corresponding to a local maximum in $G_V(r)$.



Local configuration in the vicinity of a hole of radius $r = 1.34D$, corresponding to a local minimum in $G_V(r)$.



Local configuration in the vicinity of a hole of radius $r = 1.61D$, corresponding to a shoulder in $G_V(r)$.



Local configuration in the vicinity of a hole of radius $r = 1.92D$, corresponding to a local minimum in $G_V(r)$.

- **The maximum in $G_V(r)$ at $r \sim 1.22D$ still corresponds to a highly strained 6-coordinated shell, which can be considered as a transition state between 6- and 9-coordinated configurations.**
- **The minimum in $G_V(r)$ at $r \sim 1.34D$ corresponds to $Z = 9$.**
- **The shoulder in $G_V(r)$ at $r \sim 1.61D$ corresponds to $Z = 10$.**
- **The minimum in $G_V(r)$ at $r \sim 1.92D$ corresponds to $Z = 12$.**
- **The stability of 9-, and 12-coordinated holes might be due to the fact that they better preserve the hexatic order.**

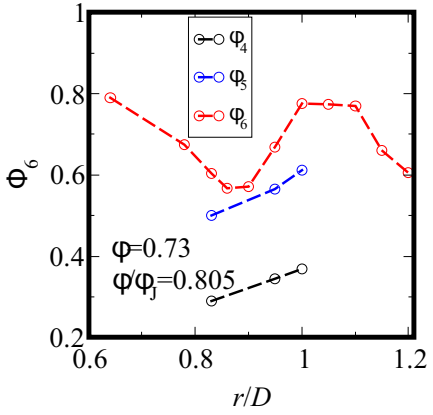
5.5 Hole geometry and bond-orientational order parameter

- **To quantitatively describe the local order in the vicinity of holes, we use the Nelson-Halperin bond-orientational order parameter Φ_6, which measures the hexatic order in a 2D state.**

$$\Phi_6 = \frac{1}{N_b} \sum_{i}^{N_b} \exp(6i\theta), \tag{9}$$

where the summation is over all N_b bonds in the configurations, and θ is the angle that bond i makes with an arbitrary fixed reference orientation, e.g., the positive x-axis. (Nelson and Halperin, PRB, 1979; Kansal, Truskett and Torquato, JCP, 2000.)

- **We consider the 18 nearest particles to the test point (about 2 coordination shells). We consider a "bond" to form between two particles if their distance is $r_b < 1.3D$.**



- **Φ_6 has a minimum at $r = 0.86 \approx \sqrt{3}/2$ and a maximum at $r = 1$: The dependence of Φ_6 on r is consistent with the trend in $G_v(R)$.**
- **$\Phi_4(r)$ and $\Phi_5(r)$ for the nearest 18 particles have the same trend in r as $\Phi_6(r)$, with $\Phi_6 > \Phi_5 > \Phi_4$. Four-coordinated holes appear induce disorder in general.**

5.6 Dependence of $G_V(r)$ on ϕ

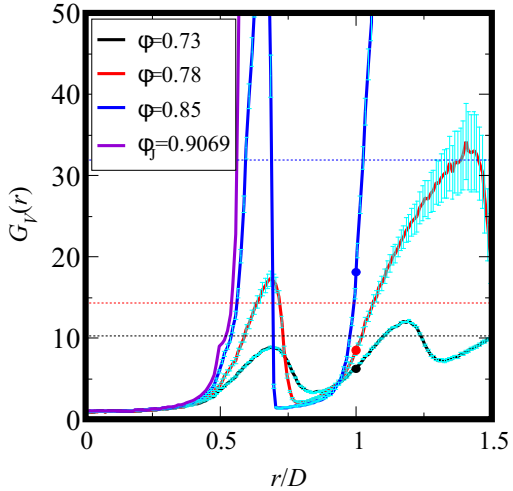


Figure: $G_V(r)$ for 2D hard disk crystals at different packing fractions. Horizontal lines: reduced pressure; Solid dots: Theoretical $G_V(D)$.

- **As ϕ increases, the first peak in $G_V(r)$ become higher and less symmetrical. The first minimum shift to smaller r, whereas the first maximum is relatively insensitive to ϕ.**

ϕ	r/D at first maximum	r/D at first minimum
0.73	0.695	0.845
0.78	0.695	0.785
0.85	0.675	0.715

- **The first maxima are roughly at $r/D = \sqrt{2}/2$, corresponding to half the diagonal of a square with side length D.**

5.6 Dependence of $G_V(r)$ on ϕ

- **In the grand canonical ensemble, there are two mechanisms to create holes slightly larger than triangular holes in a perfect triangle lattice with radius $r_t/D = \sqrt{\pi/(2\phi)}/3^{3/4}$:**
  - **(a) distortion of the triangular hole to form 4- and 5- coordinated holes;**
  - **(b) Creation of a vacancy.**

  At larger ϕ, effect (b) becomes increasingly prevalent, shifting the first minimum to smaller r.

5.7 $G_V(r)$ for a dense hard-disk liquid

To investigate whether the non-monotonicity of $G_V(r)$ is present in the liquid phase, we simulated a liquid at $\phi = 0.67 = 0.985\phi_f$, where ϕ_f is the freezing packing fraction.

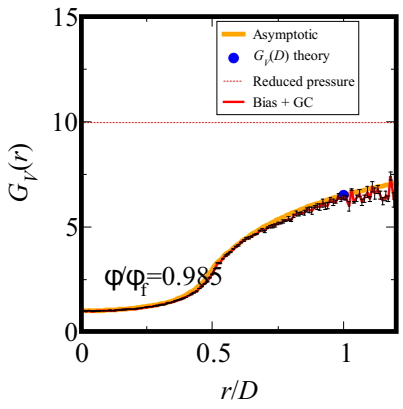


Figure: $G_V(r)$ for 2D hard disk fluid at $\phi = 0.67$, just below the freezing packing fraction.

- **The simulated $G_V(r)$ agrees very well with the asymptotic formula in [Torquato, PRE (1995)], increasing monotonically.**

$$G_V(r) = a_0 + \frac{a_1}{r}, \tag{10}$$

$$a_0 = \frac{1 + 0.128\phi}{(1-\phi)^2} \tag{11}$$

$$a_1 = \frac{-0.564\phi}{(1-\phi)^2} \tag{12}$$

- **Thus, the non-monotonicity of G_V for the crystal phase indeed results from the global orientational order, not from the coordination shells.**

Hole statistics for equilibrium FCC crystals

- **We consider a FCC crystal at $\phi = 0.55 = 1.01\phi_{\textbf{melting}} = 0.742\phi_J$.**

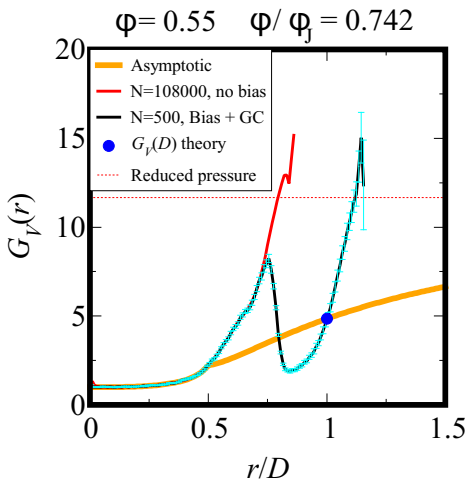


Figure: $G_V(r)$ for 3D hard sphere FCC crystal at $\phi = 0.55$, just above the melting packing fraction.

- **As in the 2D cases, biased sampling in the canonical ensemble enables one to obtain $G_v(r)$ over a much larger range than the standard sampling method.**
- **First maximum of $G_V(r)$ is at $r = 0.755D$, first minimum is at $r = 0.845D$.**
- **A small shoulder is observed at $r = 0.66$, indicating the transition from tetrahedral holes to octahedral holes.**
- **The first maximum is at larger r than those of the 2D $G_V(r)$'s due to the octahedral holes, which stabilize holes with radius $\sim \sqrt{2}/2D$.**

6. Predicting hole statistics near jamming

- **We aim to develop an analytical expression for the hole statistics in the triangle-lattice crystal near jamming.**
- **Assuming that there are no interstitials and only isolated vacancies in the crystal near ϕ_J, $H_V(r)$ can be written as**

$$H_V(r) = (1 - X)H_{V,T}(r) + XH_{V,H}(r), \tag{13}$$

  where $H_{V,T}(r)$ and $H_{V,H}(r)$ are nearest-neighbor distributions for test points in triangular and hexagonal holes, respectively, and X is the vacancy concentration.
- **X can be computed via [Pronk and Frenkel, JPCB (2001)]**

$$X = \exp(-\beta(\mu - f_1)), \tag{14}$$

  where $\exp(-\beta f_1) =: \overline{A}$ is the average area accessible to the center of a hard disk when one attempts to insert it into a vacancy [Bates and Frenkel, PRE (2000)].

6.1 Cell model

- We use a cell model in which particles are assumed to move independently within their Voronoi cells. The Voronoi tessellation is taken to be that for the perfect lattice with one single vacancy.
- Assume also that we are sufficiently near jamming that the displacement set accessible to each particle approaches its *limiting polytope* [Donev, Torquato and Stillinger, PRE (2005)].

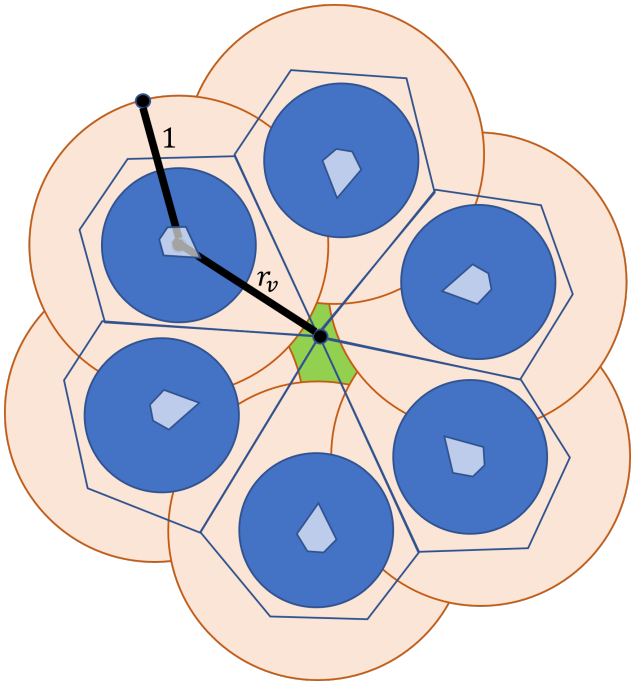


Illustration of a vacancy in the cell model. The polygons with blue edges are the Voronoi cells. The light blue regions are the accessible displacement for each particle (i.e., limiting polytope). The orange disks are the exclusion regions for each particle. The green region $\mathcal{A}$ is the region accessible to the center of an inserted disk.

6.1.1 Distribution of r_V in the cell model

- **The distribution of the distance r_v from the vacancy site to any of its 6 surrounding particles can be expressed analytically:**

$$p(r_v';\phi) \propto \begin{cases} \frac{\pi r_v'}{3}, & 0 \le r_v' < \frac{2(r_1(\phi)-1)}{\sqrt{3}} \\ \theta_1(r_v';\phi) r_v', & \frac{2(r_1(\phi)-1)}{\sqrt{3}} \le r_v' < \frac{3(r_1(\phi)-1)}{2} \\ (\theta_1(r_v';\phi) - \theta_2(r_v';\phi)) r_v', & \frac{3(r_1(\phi)-1)}{2} \le r_v' < \sqrt{\frac{7}{3}} \left| \frac{\sqrt{\frac{\pi}{2}}\sqrt{\frac{1}{\phi}}}{\sqrt[4]{3}} - 1 \right| \\ 0, & r_v' \ge \sqrt{\frac{7}{3}} \left| \frac{\sqrt{\frac{\pi}{2}}\sqrt{\frac{1}{\phi}}}{\sqrt[4]{3}} - 1 \right|, \end{cases} \tag{15}$$

where $r_v' = r_v - 1, r_1(\phi) = \sqrt{\frac{\pi}{2\sqrt{3}\phi}}$ **and**

$$\theta_1(r_v';\phi) = -2\tan^{-1}\left(\frac{\sqrt{6}\sqrt{6r_v'^2 + 2\ 3^{3/4}\sqrt{2\pi}\sqrt{\frac{1}{\phi}} - \frac{\sqrt{3}\pi}{\phi} - 6} - 3\sqrt[4]{3}\sqrt{2\pi}\sqrt{\frac{1}{\phi}} + 6\sqrt{3}}{3\sqrt{2}\sqrt{6r_v'^2 + 2\ 3^{3/4}\sqrt{2\pi}\sqrt{\frac{1}{\phi}} - \frac{\sqrt{3}\pi}{\phi} - 6} + 3^{3/4}\sqrt{2\pi}\sqrt{\frac{1}{\phi}} - 6} \right) \tag{16}$$

$$\theta_2(r_v';\phi) = 2\tan^{-1}\left(\frac{\sqrt{4r_v'^2 + 3\ 3^{3/4}\sqrt{2\pi}\sqrt{\frac{1}{\phi}} - \frac{3\sqrt{3}\pi}{2\phi} - 9}}{\frac{2}{4}\left(3^{3/4}\sqrt{2\pi}\sqrt{\frac{1}{\phi}} - 6\right)} \right) \tag{17}$$

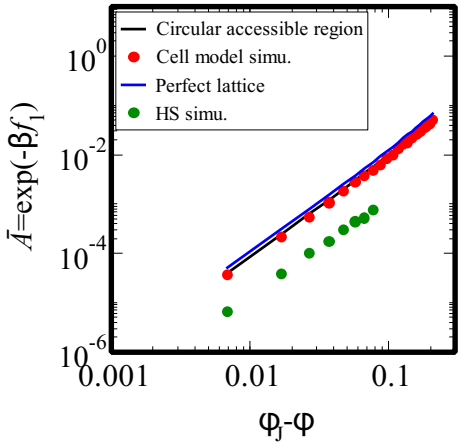
6.1.2 Accessible area a in cell model and simulations

We compute $\overline{A} = \langle |\mathcal{A}| \rangle$ for the following cases

- **Perfect triangle lattice with one vacancy: $\mathcal{A}$ is simply a regular hexagon of side length $2(r_1(\phi) - 1)/\sqrt{3}$.**
- **Circular accessible region approximation for the cell model: $\mathcal{A}$ is approximated to be circular. Thus, we have**

$$\overline{A} = \pi \langle r_v' \rangle^2, \tag{18}$$

  where $\langle r_v' \rangle = \int_{\mathbb{R}} x p(x, \phi) dx$ can computed numerically from Eq. (15).
- **Cell model simulation: $\overline{A}$ is computed directly from simulations of the cell model, averaged over 500 configurations.**
- **HS simulation: Simulation of 399 hard disks (20×20 lattice sites in a rhombic simulation box in PBC with one vacancy), averaged over 1000 configurations.**

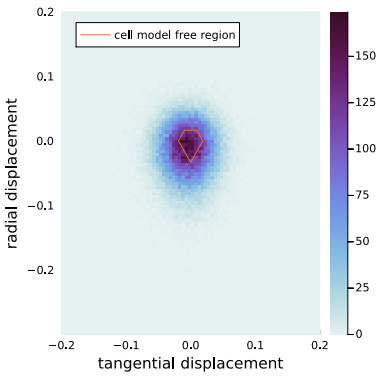


6.1.2 Accessible area a in cell model and simulations

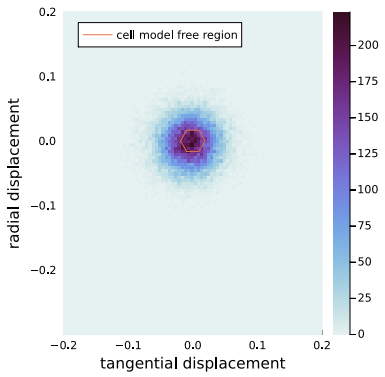
- **Near jamming, $\overline{A}$ is proportional to $(\phi_J - \phi)^2$ in all cases.**
- **In the cell model, $\overline{A}$ is smaller than that for a perfect triangle lattice.**
- **In the circular accessible region approximation, values of $\overline{A}$ agree with those in the cell model simulation within 7%.**
- **In the equilibrium hard-disk crystal, $\overline{A}$ values are 5.7 times smaller than those in the cell model simulation.**
- **The cell model seriously overestimates $\overline{A}$ because it does not allow particles to move beyond their limiting polytopes, which is much more restrictive on the particle displacements than in the equilibrium hard-disk simulation. A better model is needed to accurately predict the vacancy concentration.**

6.1.3 Particle displacement distribution in cell model and simulations

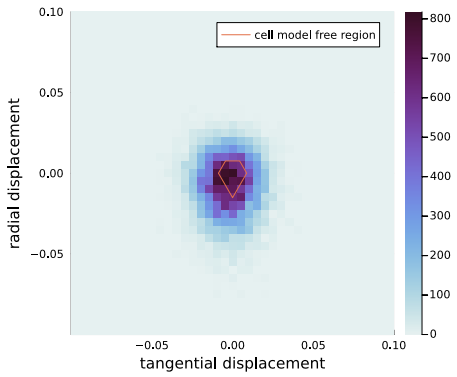
- To understand the difference between the cell model and the equilibrium HS crystal, we compute the distribution of particle displacements from their positions in the perfect lattice in a equilibrium crystal.
- Simulations are performed in a hexagonal box in periodic boundary conditions with no or one vacancy. In the case with vacancy, the vacancy is the central particle of the box.
- The center of mass is fixed at the centroid of the box.



Particle displacement distribution for the 6 particles surrounding the vacancy at $\phi = 0.85$.



Particle displacement distribution for particles in a vacancy-free crystal.



Particle displacement distribution for the 6 particles surrounding the vacancy at $\phi = 0.88$

- In the simulation of the equilibrium crystal, particles can move well beyond their Voronoi cells. The standard deviation of the displacement in the equilibrium crystal is about 3 times larger than that in the cell model.

6.2 Self-consistent-field model

- **Consider a particle p surrounded by Z particles $\mathbf{r}_1, \ldots, \mathbf{r}_Z$ in its first coordination shell.**
- **Let $f_p(\mathbf{r})$ be the pdf of the displacement of p from its position in the perfect lattice.**
- **Let $f_Z(\mathbf{r}_1, \ldots, \mathbf{r}_Z)$ be the joint pdf of the displacements of the surrounding particles from their positions in the perfect lattice.**
- **The probability that p is in the volume element $\mathbf{r}d\mathbf{r}$ is proportional to the probability that the distances from p to all the Z particles are greater than D:**

$$f_p(\mathbf{r}) \propto \int_{\{\mathbf{r}_1, \ldots \mathbf{r}_Z\} \notin \mathcal{B}(\mathbf{r}, D)} f_Z(\mathbf{r}_1, \ldots, \mathbf{r}_Z) d\mathbf{r}_1 \ldots d\mathbf{r}_Z. \tag{19}$$

- **The joint pdf f_Z is estimated to be the product of pdf's for all Z surrounding particles, subject to the constraint that no two of the Z particles overlap, i.e.,**

$$f_Z(\mathbf{r}_1, \ldots, \mathbf{r}_Z) \propto \begin{cases} 0, & |\mathbf{r}_j - \mathbf{r}_k| < D \text{ for any } j, k \in \{1, \ldots, Z\} \\ \prod_j f_{p,j}(\mathbf{r}_j), & \text{otherwise}. \end{cases} \tag{20}$$

- **Equations (19) and (20) form self-consistent relations that can be solved numerically.**

6.2 Self-consistent-field model

We make the following assumptions to simplify calculations.

- $f_p(\mathbf{r})$ **can be approximated by a multivariate** d**-dimensional normal distribution.**
- $f_Z(\mathbf{r}_1, \ldots, \mathbf{r}_Z)$ **can be approximated by a** dZ**-dimensional multivariate normal distribution.**

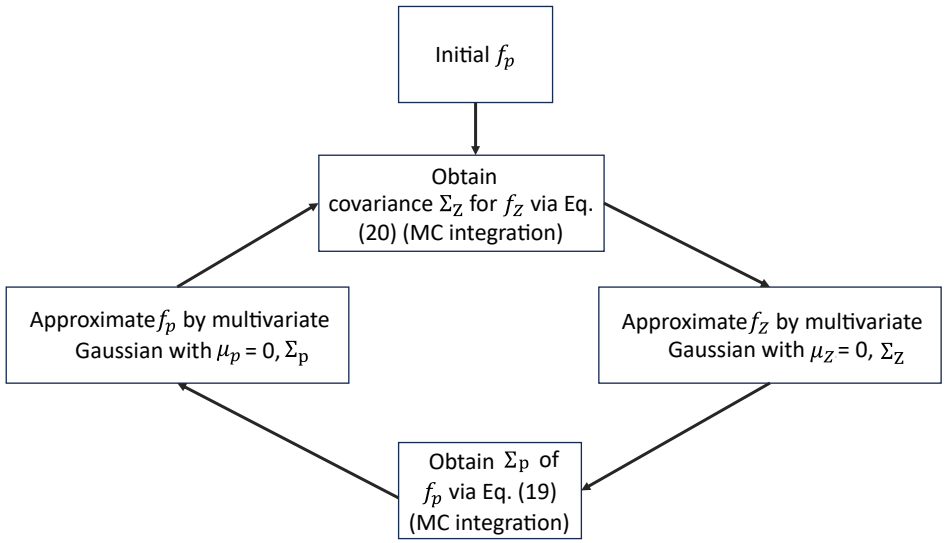
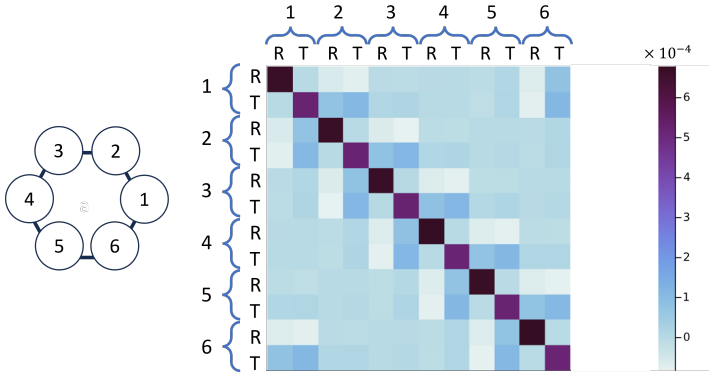


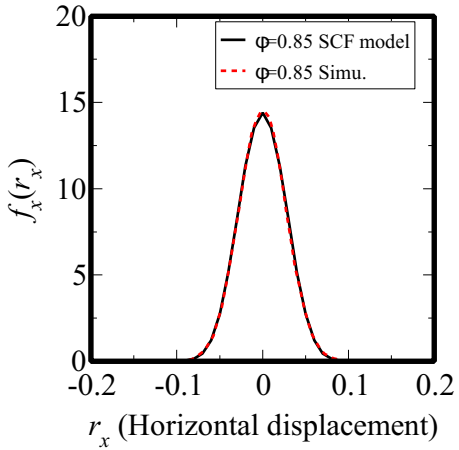
Figure: Flowchart of the SCF algorithm.

- **The iterations stop when** $f_{p,i+1}$ **and** $f_{p,i}$ **differ within 2% at their maxima.**

6.2.1 SCF model for a vacancy-free triangle-lattice crystal



Covariance matrix Σ_Z for the displacements of the 6 nearest neighbors of a particle in the 2D triangle-lattice crystal at $\phi = 0.85$ obtained via the SCF model. The labels 1–6, R and T refer to the radial and tangential components of the 6 particles.

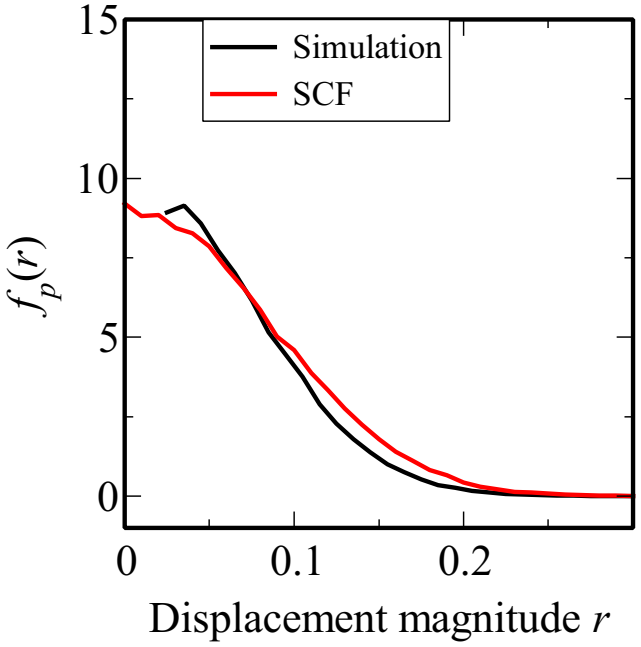


Marginal pdf $\int_{\mathbb{R}} f_p(r_x, r_y) dr_y$, i.e., pdf of horizontal displacements for 2D hard-disk systems obtained via the SCF model and simulation.

- The correlation in displacements of the surrounding particles is almost exclusively due to neighboring ("ortho") particles. Meta- and para-positioned particles can be treated as independent.
- The correlation matrix shows that particles 1, 3, 5 and 2, 4, 6 tend to move in opposite radial directions. Thus, the central particle's actual displacement distribution is more restricted than in the case where particles 1–6 are mutually independent.
- Both the simulated displacement distribution and the distribution via the SCF model can be very well approximated by Gaussians. The decay is still Gaussian-like for very small systems, but with a smaller standard deviation.
- The SCF model well approximates the displacement pdf in 2D vacancy-free HS crystals. At $\phi = 0.85$, the discrepancy in standard deviation between SCF model ($0.0278D$) and simulation ($0.0272D$) is 2%. The SCF works well in the entire crystal density range $0.73 \leq \phi \leq \pi/\sqrt{12}$.

SCF model for a vacancy-free fcc crystal

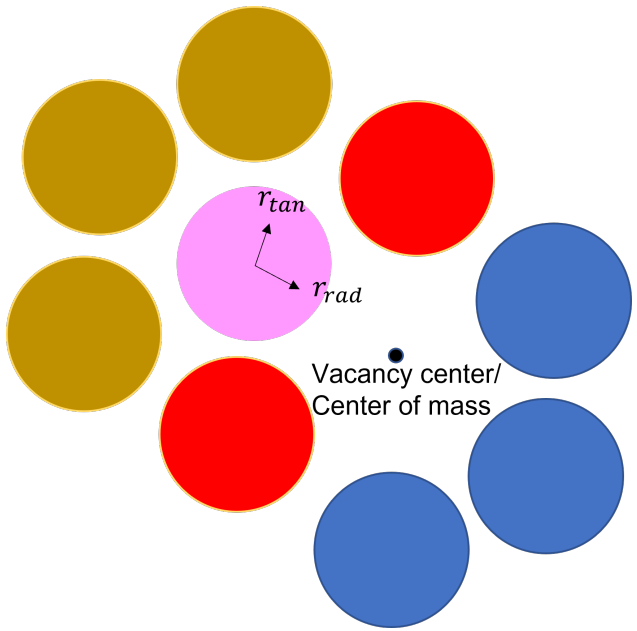
- **For the 3D fcc crystal, the SCF model predicts the simulated displacement distribution with good accuracy. However, the agreement is worse than for the the 2D crystal.**
- **The SCF model overestimates the spread of the distribution because only the first coordination shell is considered.**
- **Use a fixed second coordination shell?**



Displacement distribution $f_p(\mathbf{r})$ along the (1, 0, 0) direction of a fcc crystal at $\phi = 0.55$, i.e., $\mathbf{r} = [r, 0, 0]^T$.

6.2.2 SCF model for the triangle lattice with one vacancy

- **Now we consider the triangle-lattice crystal with one single vacancy. Each particle (e.g., the pink particle) surrounding the vacancy now have $Z_1 = 5$ neighbors in its first coordination shell (brown and red particles).**
- **The displacement pdf of the brown particles are fixed to be the one for the vacancy-free crystal, because they have 6 neighbors. The displacement pdf of the red and pink particles are solved self-consistently.**



6.2.2 SCF model for the triangle lattice with one vacancy

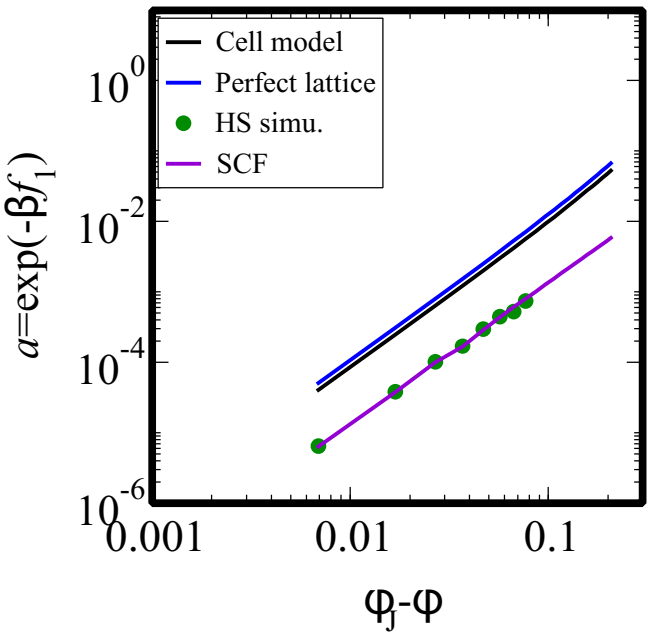
- **We consider** $\phi = 0.85$**.**
- **The inward radial displacement** r_{rad} **is** $(0.016 \pm 0.042)D$ **via the SCF model, and is** $(0.016 \pm 0.038)D$ **via simulation. The SCF model overestimates the variance of particle displacements by 10%, but can be considered a good approximation.**
- **The distance between opposite particles** $d_{\leftrightarrow}$ **around the hexagonal vacancy is** $(2.033 \pm 0.059)D$ **via the SCF model, and is** $(2.034 \pm 0.054)D$ **via hard-disk simulation.**
- **The accessible region** $\mathcal{A}$ **on p. 23 vanishes as long as one of the three** $d_{\leftrightarrow}$ **is** $< 2D$**. Thus, we use the smallest of the three** $d_{\leftrightarrow}$ **to compute the accessible region area.**

$$\overline{A} = \pi \left[\max \left(0, \min_{i=1}^{3} \frac{d_{\leftrightarrow,i} - 2D}{2} \right) \right]^2 \tag{21}$$

- **The SCF model yields** $\overline{A} = 0.00044$**, very close to** $\overline{A} = 0.000445$ **via simulation.**

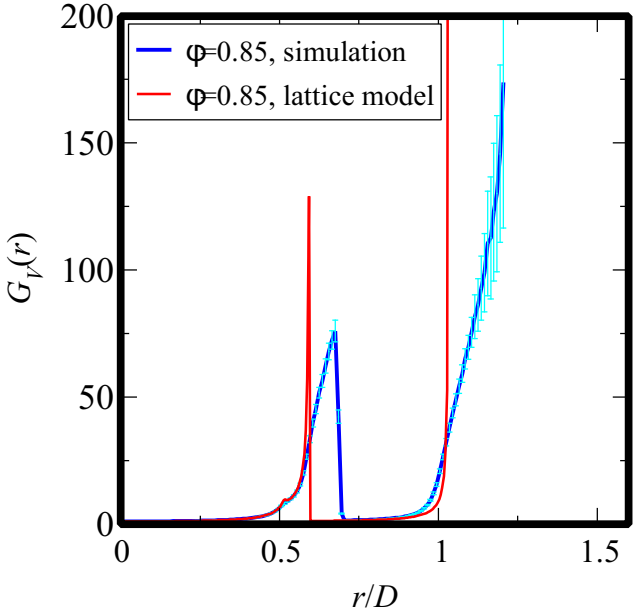
6.2.2 SCF model for the triangle lattice with one vacancy

The SCF model provides a very good estimate of $\overline{A}$ and thus of the vacancy concentration via (14).



6.3 Hole statistics in the lattice model

Consider a lattice model, where particles can hop from site to site but do not vibrate. The triangular holes and hexagonal holes are then perfect regular polygons (no distortions).
Then $G_V(r)$ can be computed via Eq. (13)–(14), together with formulas for $H_V(r)$ for perfect triangular and hexagonal holes [Torquato, PRE 2010] as well as the vacancy concentration obtained via the SCF model.

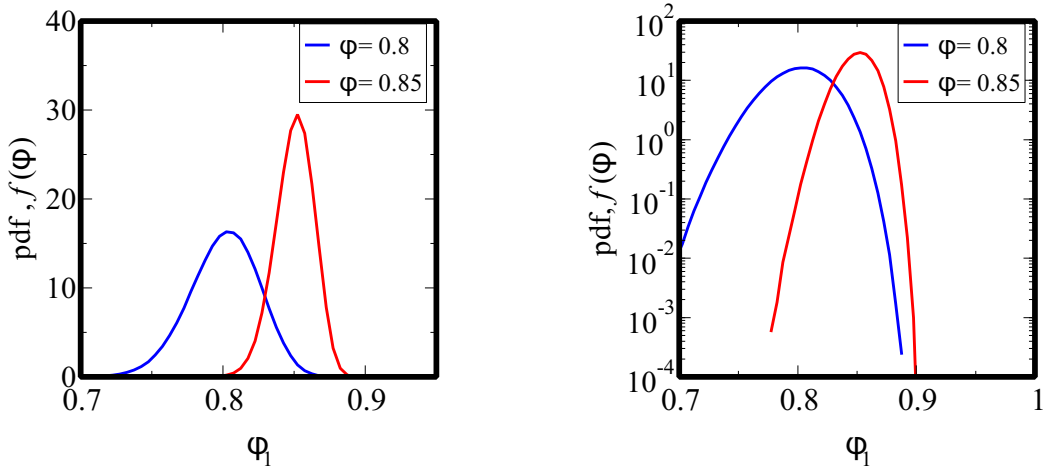


The lattice model predicts the existence of the first two peaks in $G_V(r)$. The predicted peaks are narrower than simulations because the lattice model does not consider distortions of the holes.

7. Local Packing Fraction Statistics I

We compute the probability distribution $f(\phi_1)$ of the local packing fraction of the first coordination shell for any particle in a hard-disk crystal. For the triangle lattice, this quantity is given by

$$\phi_1 = \frac{3\pi D^2}{\textbf{Area of the hexagon formed by the first coordination shell}} \tag{22}$$



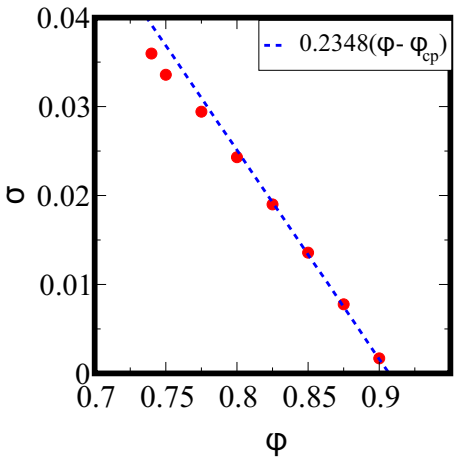
The decay of $f(\phi_1)$ of ϕ_1 is Gaussian-like. However, the distribution is asymmetric. The tail towards smaller ϕ_1 is wider, because ϕ_1 is bounded above by the close-packing fraction $\phi_{\mathbf{cp}} = \pi/\sqrt{12}$.

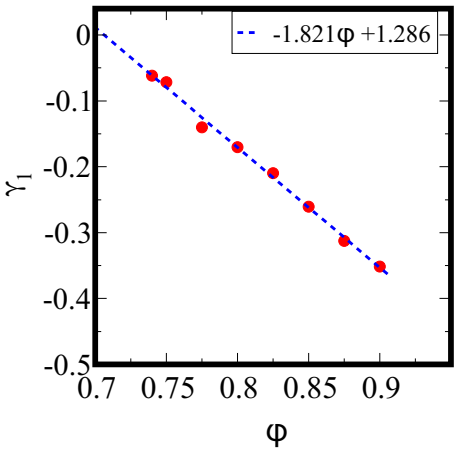
7. Local Packing Fraction Statistics II

We compute the standard deviation σ and the skewness γ_1 for the local packing fraction distributions. The skewness is defined as

$$\gamma_1 = \frac{E[(X-\mu)^3]}{E[(X-\mu)^2]^{3/2}}, \tag{23}$$

where $\mu = \phi$, the total packing fraction. We can compute $f(\phi_1;\phi)$ for $\phi \geq 0.74$. Below this packing fraction, particles swap frequently, and the first coordination shell becomes ill-defined.





7. Local Packing Fraction Statistics III

- **Both σ and γ_1 vary linearly with ϕ.**
- $\gamma_1(\phi) < 0$**, showing that** $f(\phi_1; \phi)$ **are left-skewed.**
- **Interestingly, The trendline for** $\gamma_1(\phi)$ **reaches zero at** $\phi = 0.70$**, approximately the melting packing fraction.**
- $|\gamma_1|$ **reaches maximum at close packing:** $\gamma_1(\phi \to \phi_{\mathbf{cp}}) \to -0.366$**. It will be instructive to compare this limit with 1D and 3D cases.**

Conclusions I

- **The biased sampling scheme in the grand canonical ensemble is able to sample rare large holes that are nearly impossible to observe in the standard method. Thus, it provides an accurate algorithm to compute the hole statistics up to hole radii such that $E_V(r) = 10^{-50}$.**
- **The biasing sampling scheme clearly shows that $G_V(r)$ is oscillatory for the 2D triangle lattice. The first three minima in $G_V(r)$ corresponds to 6-, 9- and 12-coordinated holes.**
- **The dependence of the bond-orientational order parameter Φ_6 on hole radius r is consistent with the trend in $G_V(r)$. Thus, hole sizes that preserve local hexatic symmetry of the triangle lattice are favored.**
- **G_V is monotonic for dense liquids at ϕ just below freezing. The non-monotonicity of G_V results from global orientational order, not from coordination shells.**
- **The SCF model shows that multivariate normal distributions very well describes the displacement distribution of particles.**
- **The skewness of the local packing fraction for hard disks varies linearly with ϕ in almost the entire crystal range, and vanishes at around the melting packing fraction.**